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Code : 303204

BCA 2nd Semester Exam., 2018

PROBLEM SOLVING TECHNIQUE AND PROGRAMMING IN C

Time : 3 hours *Full Marks* : 60

Instructions :

(i) All questions carry equal marks.

(ii) There are ***SEVEN*** *questions in this paper.*

(iii) Attempt ***FIVE*** *questions in all.*

(iv) Question Nos. ***1*** *&* ***2*** *are compulsory.*

1. Choose the correct answer of the following (any *six*) :

(a) In which header file is the NULL macro defined?

(i) stdio.h

(ii) stdder.h

(iii) stdio.h and stddef.h

(iv) math.h

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(b) If a variable is a pointer to a structure, then which of the following operators is used to access data members of the structure through the pointer variable?

(i) .

(ii) &

(iii) *

(iv) – >

(c) Which is the stage of the following code?

#include<stdio.h>

gets replaced by the contents of the file stdio.h

(i) During editing

(ii) During linking

(iii) During execution

(iv) During preprocessing

(d) How will you free the allocated memory?

(i) remove(var-name);

(ii) free(var-name);

(iii) delete(var-name);

(iv) dalloc(var-name);

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(e) What is the similarity between a structure, union and enumeration?

(i) All of them let you define new values

(ii) All of them let you define new data types

(iii) All of them let you define new pointers

(iv) All of them let you define new structures

(f) What does the following declaration mean?

int (*ptr)[10];

(i) ptr is array of pointers to 10 integers

(ii) ptr is a pointer to an array of 10 integers

(iii) ptr is an array of 10 integers

(iv) ptr is a pointer to array

(g) When is the 'void' keyword used in a function? CSVTUonline.com

(h) How do you generate random numbers in C?

(i) What are actual arguments?

(j) What are formal parameters?

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2. Answer any *three* of the following :

(a) What is an endless loop?

(b) What are comments and how do you insert it in a C program?

(c) What is variable initialization and why is it important?

(d) What is pointer to a function?

(e) What is type casting?

3. What are command line arguments of main function? Explain with a suitable example.

4. What are header files? Are functions declared or defined in header files?

5. What is a pointer in C? What are the advantages of using pointers? Explain with examples.

6. What is a structure? Explain structure within structure with an example.

7. What are the differences between formal arguments and actual arguments of a function? Explain with an example.

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